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Project Title: AI-Based Intelligent Solution for Healthcare & Well-being

stressina.AI – A Fuzzy Logic Approach to Real-Time Stress Classification and Adaptive Coping Recommendation

FINAL REPORT

Group : 12

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Project repository: https://github.com/h1rum4l/ai-wellbeing-project

Live prototype: https://stressinaai-neon.vercel.app

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Abstract

Most of us only realise how stressed we are when we are already worn out. The usual ways of measuring stress, like questionnaires and counselling sessions, depend on people remembering how they felt, and they cannot tell us anything about the present moment. In this project we tried to build a simple tool that gives an early estimate of stress using three everyday indicators: Heart Rate Variability (HRV), daily screen time and sleep duration.

The system, which we named stressina.AI, has two main parts. The first is a Mamdani-type Fuzzy Inference System that takes the three inputs and produces a Stress Index from 0 to 100 using 31 IF–THEN rules and centroid defuzzification. The second is a separate rule-based engine that chooses a coping suggestion depending on the stress category and on how much time and what setting the user has. An explanation module shows the membership values and fired rules so that the result is never a black box. In Stage 2 we built this as a Python and Streamlit prototype. For the final stage we rebuilt the interface as a web application using React, TypeScript, Vite and Tailwind CSS with a "Liquid Glass" design, and added a breathing and relaxation area, a panda mascot companion called Stressina, a personal history page with charts, and local account and privacy controls.

The fuzzy logic was checked with an automated pytest suite (26 test functions, 71 cases) and with a 122-row synthetic dataset. For an average workday (HRV 65 ms, 6 hours of screen time, 7 hours of sleep) the system gave a Stress Index of 35.5, which is Low Stress. Since the data is simulated and the rules were designed by us, the tool is an educational prototype and not a medical device. The report also discusses its limitations and the ethical points we considered.

Keywords: fuzzy logic, Mamdani inference, stress classification, heart rate variability, explainable AI, decision support, well-being.

1. Introduction

Stress has become part of daily life for many university students and office workers. Deadlines, long hours in front of a screen and short nights of sleep all pile up, and the effect shows in poor concentration, tiredness and sometimes health problems. The difficulty is that there is no easy way to check our stress level on an ordinary day, so most people do nothing until the situation gets serious.

This report presents the final stage of our IT3182 group project, "AI-Based Intelligent Solution for Healthcare & Well-being". In Stage 1 we proposed the idea and the design. In Stage 2 we

reported the membership functions, the rule base and the Python back-end. In this final report we put everything together, including the changes we made afterwards: the application was renamed stressina.AI and its user interface was redesigned and extended.

The rest of the report is organised as follows. Sections 2 and 3 explain the background and the problem. Section 4 lists the objectives. Sections 5 and 6 describe and justify the AI techniques. Sections 7 and 8 cover the design and implementation. Section 9 gives testing and results, Section 10 discusses limitations, Section 11 covers ethical and social issues, and Section 12 concludes.

2. Problem Background

Long-term psychological stress is often called a silent epidemic in schools, universities and workplaces. It is linked with anxiety, heart problems and weaker thinking and memory [3], [8]. Students have assignments, exams and presentations, while office staff sit at computers for most of the day with few breaks. For both groups, irregular sleep and many hours on screens make things worse.

Stress is normally assessed through clinical consultations or self-reported questionnaires such as the 10-item Perceived Stress Scale. These methods are well validated, but they are subjective, they need effort from the person, and they only describe what already happened. Because of this, people often notice their stress only after reaching burnout.

Research on wearable sensors and machine learning shows that physiological signals, especially HRV, contain useful information about stress [4], [5], [6]. However, many of these systems work like black boxes and do not explain their reasoning to an ordinary user. So there is a need for a low-effort and understandable tool that can read simple daily indicators and suggest practical ways to cope.

3. Problem Statement

Existing well-being tools either depend on invasive or rigid biological measurements, such as medical-grade cortisol tests, or apply fixed thresholds that ignore context. Body signals are naturally vague: a small change in a reading does not suddenly turn a relaxed person into a stressed one. Current tools do not handle this uncertainty well, and they cannot explain how they reached a result or give personalised advice.

The core problem for this project is therefore how to classify stress from non-invasive indicators in a way that respects uncertainty, can be explained step by step, and leads to useful, context-aware recommendations.

4. Objectives

The aim of the project is to design and build an explainable AI-based decision-support prototype for everyday stress awareness. The specific objectives are:

- To classify stress using a Mamdani-type Fuzzy Inference System driven by HRV, daily screen time and sleep duration, giving a Stress Index (0–100) and a Low, Medium or High category.
- To recommend coping actions through a separate rule-based decision-support layer that considers the stress category together with the user's available time and setting.
- To build a working interactive prototype that shows the reasoning process clearly, including membership degrees and fired rules. This started as a Python/Streamlit desktop simulator and was later developed into the stressina.AI web application.
- To validate the system using automated tests and a synthetic dataset.

5. Selected AI Techniques

5.1 Mamdani Fuzzy Inference System

Fuzzy logic, introduced by Zadeh [1], lets a value belong to a set partly, with a degree between 0 and 1, instead of being fully in or fully out. In the Mamdani approach [2], knowledge is written as linguistic rules such as "IF HRV is Low AND Screen Time is Prolonged AND Sleep is Inadequate THEN Stress is High". The system works in four steps: fuzzification of the crisp inputs, evaluation of the rules (min for AND), aggregation of the rule outputs (max), and defuzzification to a single number. We used the centroid method for defuzzification.

5.2 Rule-Based Reasoning

After the stress category is known, the system has to decide what to suggest. For this we used a deterministic rule-based decision-support component. Its rules link the stress category, the time the user has (5, 10 or 20+ minutes) and the setting (Home, University, Work or Other) to one action from a fixed library, for example Active Cardio Break, Breathing Exercise, Sleep Hygiene Tip, Digital Detox Reminder, Short Walk Break, Maintain Routine or No Action Needed.

5.3 Explanation Layer

To avoid a black-box result, we added an explanation module. For every assessment it reports the membership degree of each input in every linguistic term, the rules that fired and how strongly, the main stress drivers and the reason behind the chosen recommendation.

6. Justification of the AI Techniques

Fuzzy logic suits this problem because the inputs have no sharp boundaries. An HRV of 49 ms and one of 51 ms describe almost the same person, but a fixed threshold would put them in different classes. With overlapping membership functions the Stress Index changes smoothly when the inputs change, so there are no sudden jumps.

We chose a Mamdani system instead of a data-driven model for three reasons. First, the rules are written in plain language, so anyone can read and discuss them. Second, it does not need a large labelled stress dataset, which we did not have and which would also bring privacy concerns. Third, every output can be traced back to specific membership values and rules, which answers the lack of explainability described in the problem statement.

We also kept the recommendation engine separate from the fuzzy classifier on purpose. The stress score should depend only on the body and behaviour indicators, while the advice can reasonably depend on practical things like how many minutes the user has. Keeping them apart means the user's context can never change the stress score; it can only choose between actions that already belong to the assigned category. Rule-based advice is also predictable and easy to audit, which matters for a well-being tool where surprising suggestions would be a problem.

7. System Design

7.1 Inputs and Outputs

Variable	Name	Range	Linguistic terms
Input 1	Heart Rate Variability (HRV)	20–120 ms	Low, Normal, High
Input 2	Daily Screen Time	0–16 hours	Short, Moderate, Prolonged
Input 3	Sleep Duration	2–10 hours	Inadequate, Marginal, Optimal
Output 1	Stress Index	0–100	Low, Medium, High
Output 2	Coping Recommendation	Text	Action chosen from the coping library

Table 1. System inputs and outputs

7.2 System Architecture

The system follows the eight-stage pipeline proposed in Stage 1: (1) user input, (2) preprocessing, (3) fuzzification, Mamdani inference and defuzzification, (4) rule-based recommendation engine, (5) recommendation output, (6) explanation and transparency, (7) results display, and (8) continuous update, where the pipeline simply runs again whenever new values are entered. The fuzzy engine and the recommendation engine are kept as independent parts so that each can be tested and changed on its own. The web application also includes an interactive walkthrough of these eight stages.

7.3 Membership Functions

Trapezoidal and triangular membership functions were used. The final parameters are given below.

Variable	Term	Shape	Parameters
HRV (ms)	Low	Trapezoidal	[20, 20, 35, 50]
HRV (ms)	Normal	Triangular	[35, 55, 75]
HRV (ms)	High	Trapezoidal	[60, 80, 120, 120]
Screen Time (h)	Short	Trapezoidal	[0, 0, 2, 5]
Screen Time (h)	Moderate	Triangular	[3, 6, 9]
Screen Time (h)	Prolonged	Trapezoidal	[7, 11, 16, 16]
Sleep (h)	Inadequate	Trapezoidal	[2, 2, 4, 5.5]
Sleep (h)	Marginal	Triangular	[4.5, 6, 7.5]
Sleep (h)	Optimal	Trapezoidal	[6.5, 8, 10, 10]
Stress Index	Low	Triangular	[0, 0, 40]
Stress Index	Medium	Triangular	[25, 50, 75]
Stress Index	High	Triangular	[60, 100, 100]

Table 2. Membership function parameters

After defuzzification, the crisp Stress Index is turned into a category using fixed thresholds: Low below 40, Medium from 40 up to (but not including) 70, and High from 70 upwards.

7.4 Rule Base

The rule base has 31 rules in two layers. Layer A contains 27 core rules that cover every combination of the three inputs (3 × 3 × 3), so nothing is missing and nothing is repeated. Four of them (R1, R2, R3 and R8 from the Stage 1 proposal) were kept exactly as originally written. The rest were created with a simple heuristic:

- Stress-associated terms: Low HRV, Prolonged Screen Time and Inadequate Sleep.

- Positive terms: High HRV, Short Screen Time and Optimal Sleep.
- Neutral terms: Normal HRV, Moderate Screen Time and Marginal Sleep.
- High Stress is given when at least two stress-associated terms are present.
- Low Stress is given when there is no stress-associated term and at least two positive terms.
- Medium Stress covers all the remaining mixed or single-factor cases.

Layer B has four extra two-antecedent rules (R4 to R7) carried over from Stage 1. For example, R4 says that Low HRV together with Inadequate Sleep means High stress whatever the screen time, and R6 says that High HRV together with Optimal Sleep means Low stress.

7.5 Recommendation Engine Design

The recommendation engine receives only the stress category and two optional values, the available time and the setting. Each action in the library is scored against the requested duration, setting and, when available, the main stressor, and the best match is returned with a short explanation. A hard rule in the code stops the context from changing the category.

7.6 User Interface Design

In Stage 2 the interface was a four-tab Streamlit dashboard with a plain dark theme. For the final version we redesigned it completely. The new interface uses a calm "Liquid Glass" theme, with soft translucent panels, and is responsive so it works on both laptops and phones. We also gave the project a name and a friendly identity: the application is called stressina.AI and has a panda mascot, Stressina, who acts as a companion. The idea behind this change was that a stress tool should itself feel calming and approachable, and that users are more likely to come back to something that does not look like a clinical form.

Feature	Description
Real-Time Assessment	Sliders for HRV, screen time and sleep with instant re-evaluation, a stress gauge, category badge, fuzzification table and active rules breakdown
Relax & Breathing Sanctuary	Guided breathing (Box 4-4-4-4, 4-7-8 Relaxing, Coherent 5-5) with an animated circular timer and phase indicators; 5-4-3-2-1 sensory grounding; a calm card-matching game; and a mindfulness check-in
Stressina Companion	A floating panda companion with a popup panel and a quick launcher that works on mobile
My History & Analytics	Assessment history linked to the user's account, trend charts of the Stress Index over time, CSV export and conversation review
Interactive Membership Functions	Interactive SVG curves for HRV, screen time, sleep and stress output, with a marker for the current input and the centroid result

Feature	Description
System Architecture Pipeline	Interactive walkthrough of the 8-stage pipeline
Dataset Evaluation	Batch evaluation of the 122 synthetic samples, showing accuracy against the reference classes and the stress distribution
Account & Privacy Controls	Local registration and login, password hashing and privacy toggles

Table 3. Main features of the stressina.AI interface

8. Implementation Details

8.1 Tools and Technologies

The project was built in two steps. The Stage 2 prototype used Python 3.10+, scikit-fuzzy for the membership functions and inference, Streamlit for the interface and pytest for testing. For the final version the application was rebuilt as a web app with the tools below.

- React 19 and TypeScript for the user interface and the application logic.
- Vite as the build tool and development server.
- Tailwind CSS for styling and the Liquid Glass theme.
- SVG graphics for the membership-function plots, the gauge, the breathing timer and the trend charts.
- Git and GitHub for version control and teamwork, and Vercel for hosting the live version.

8.2 Project Structure

The project is stored in a public GitHub repository. Its main parts are the src folder (application source code), the data folder (the 122-row synthetic dataset), the public and assets folders (images, logo and fonts), and the configuration files for Vite, TypeScript and Vercel. The README and QUICKSTART files explain how to install and run the project.

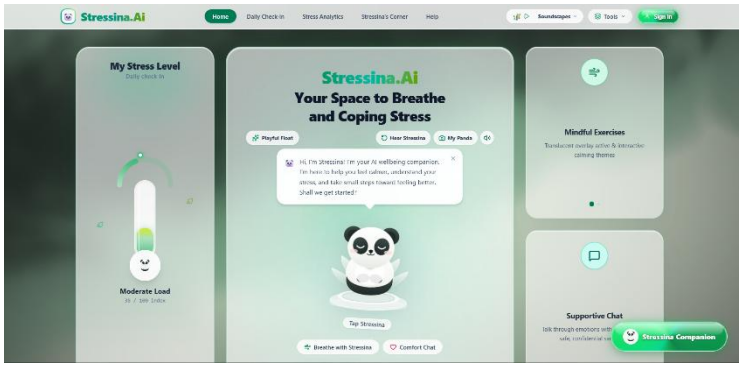
To run it locally, a user installs the dependencies with npm install and then starts the application with npm run dev. A production build is created with npm run build. The same repository is deployed online as the live prototype.

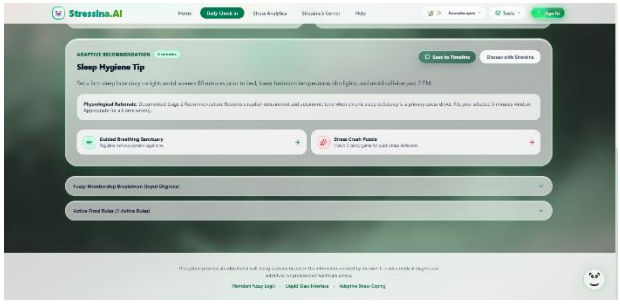
8.3 Fuzzy Engine in the Web Application

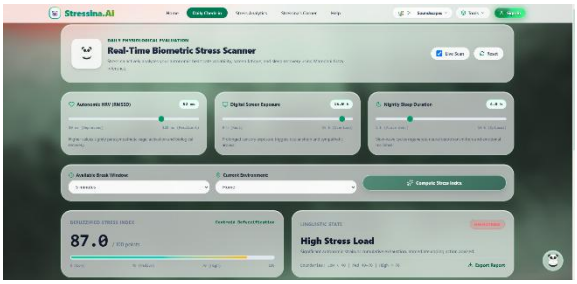
The web version keeps the same fuzzy design as Stage 2: the same input ranges, the same membership functions, Mamdani min/max inference and centroid defuzzification to a Stress Index between 0 and 100, followed by the Low, Medium and High thresholds. The

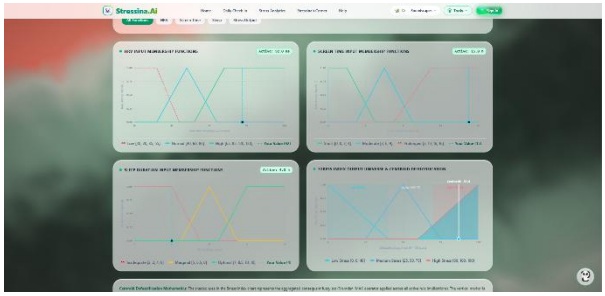
recommendation engine and the explanation layer also work as before, so the "why" behind every result can still be inspected through the active rules and fuzzification table.

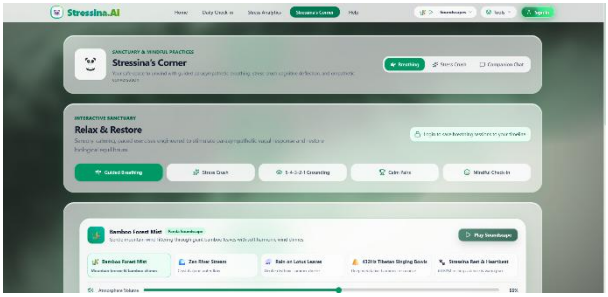
8.4 Screenshots

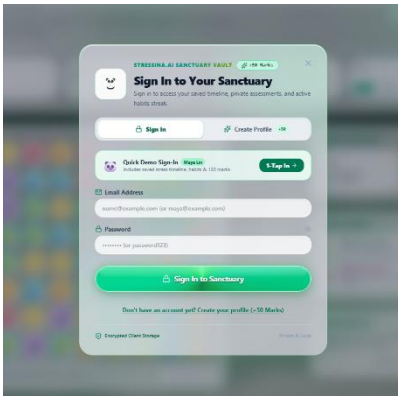


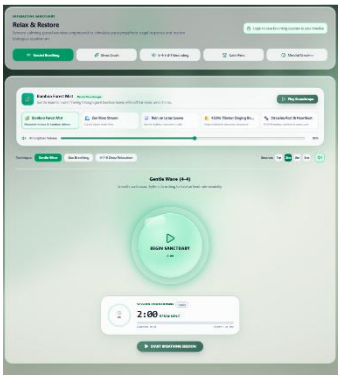












9. Testing and Results

9.1 Automated Tests

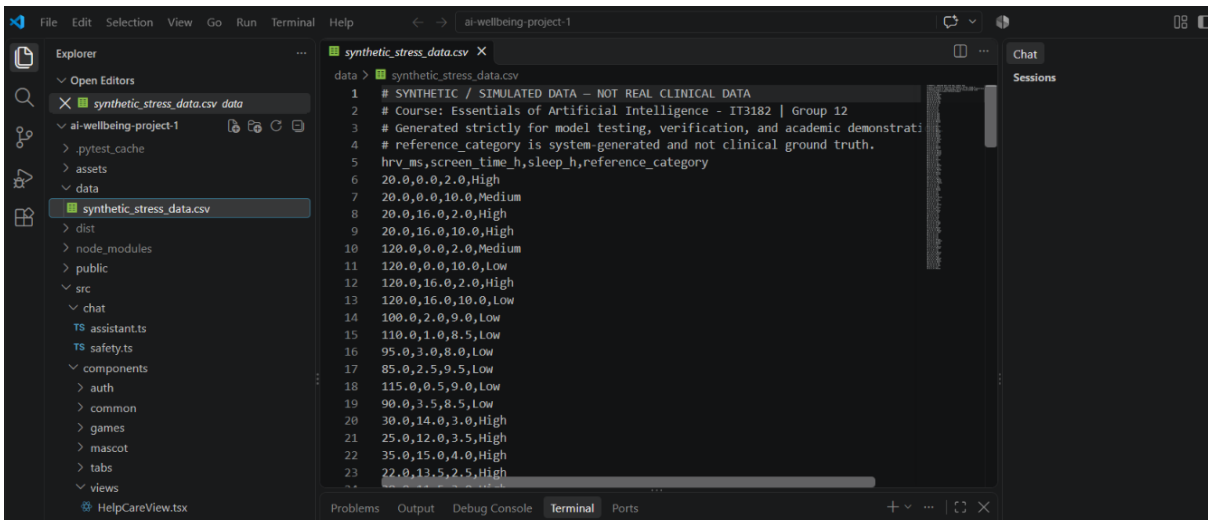
During Stage 2 an automated pytest suite was written to check the fuzzy engine, the recommendation engine and input handling. There are 26 test functions, which expand to 71 test cases after parametrisation.

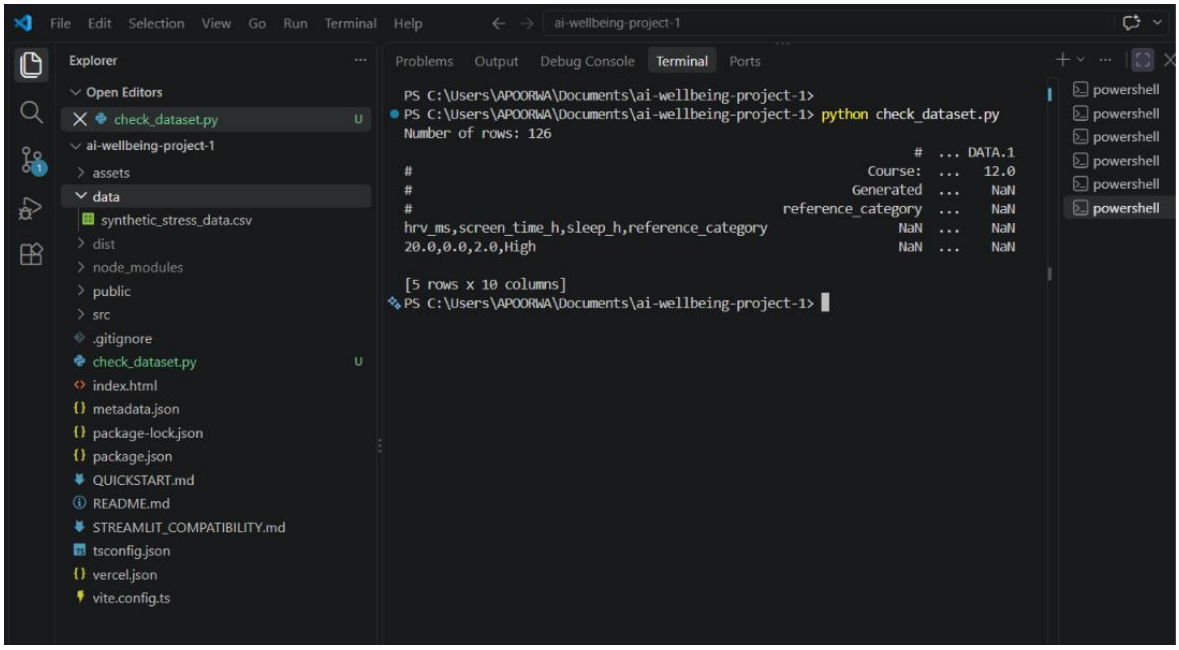
Test module	Functions	What it checks
test_fuzzy_engine.py	13	That all 27 core rules exist exactly once; that the 4 supplementary rules work on their own; documented Low/Medium/High scenarios; classification thresholds at 0, 20, 39.999, 40, 50, 69.999, 70, 85 and 100; and all 8 corner cases of the input space
test_recommendations.py	6	The full combination of stress category, available time and setting; confirms that context never changes the stress category
test_validation.py	7	Handling of out-of-range or badly formed inputs

Table 4. Automated test suite summary (Python version)

9.2 Synthetic Dataset and Batch Evaluation

A 126-row synthetic dataset was created to check the system in bulk. It contains boundary extremes, typical student and employee profiles, points where linguistic terms meet, and random samples made with a fixed seed (42) so the results can be repeated. The file is clearly marked as synthetic data and not real clinical data. Its reference category column was produced by the system's own fuzzy pipeline, so it is useful for checking that the system behaves consistently, but it is not independent ground truth and cannot show real-world accuracy. The web application has a Dataset Evaluation page that runs all 126 samples and shows the accuracy against the reference classes and the distribution of stress levels.



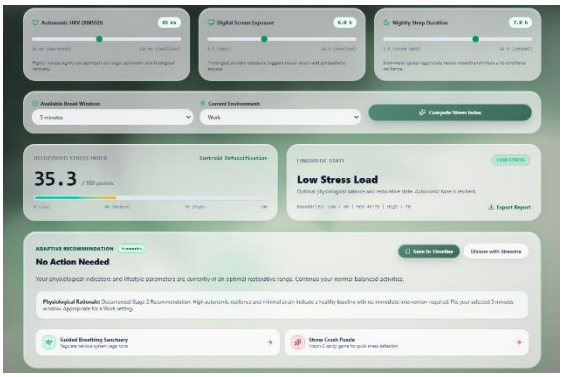


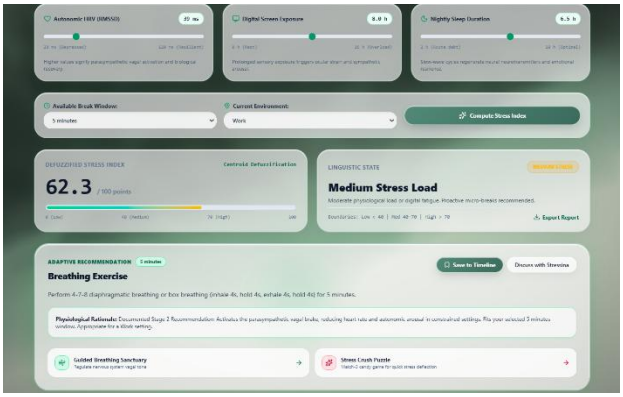
9.3 Example Result

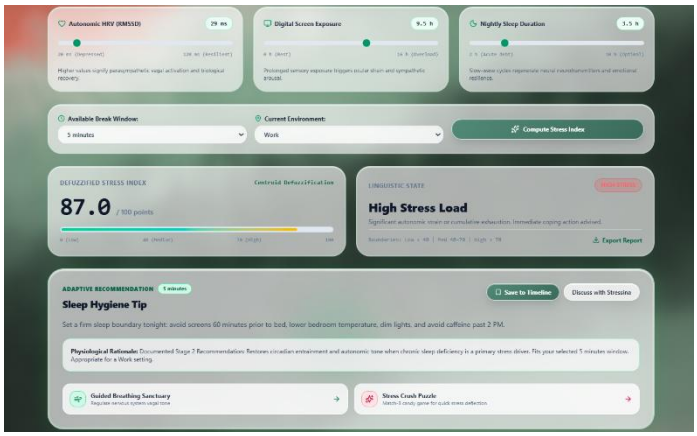
As a typical case, an average workday was entered: HRV = 65 ms, Daily Screen Time = 6.0 hours and Sleep Duration = 7.0 hours. Fuzzification gave a membership of 1.00 for Moderate screen time, about 0.33 for both Marginal and Optimal sleep, and 0.50 for Normal HRV. The defuzzified Stress Index was 35.5 out of 100, which is below 40 and so is classified as Low Stress. The recommendation engine returned "No Action Needed" with an explanation that the indicators show a healthy baseline and no immediate intervention is needed.

Input	Value	Stress Index	Category / Recommendation
HRV 65 ms, Screen 6.0 h, Sleep 7.0 h	Average workday	35.5	Low / No Action Needed
HRV 39 ms, Screen 8.0 h, Sleep 6.5 h	Average workday	62.3	Medium / Breathing exercises are recommended
HRV 29 ms, Screen 9.5 h, Sleep 3.5 h	Average workday	87	High / Significant autonomic strain or cumulative exhaustion. Immediate coping action advised

Table 5. Example assessment







9.4 Observations

When we moved the sliders, the Stress Index changed gradually instead of jumping, which shows that the overlapping membership functions are working as intended. The transparency panels also made it much easier to find mistakes while building the rule base, because we could see which rules fired for any input. The redesigned interface made the results easier to read than the earlier Streamlit version, especially the gauge and the trend charts.

10. Limitations and Discussion

The project met its main objectives, but there are some limitations we want to state honestly.

- Simulated data: we built and checked the system with synthetic data only. We did not have real physiological recordings or true stress labels, so we cannot claim real-world accuracy.

- Expert-designed parameters: the membership functions and the extra core rules were designed by our team using the heuristic in Section 7.4 and the literature. They have not been validated by psychologists or clinicians.
- Manual input: the user types in HRV, screen time and sleep. HRV in particular is hard to measure without a wearable device, so the tool is not yet truly continuous or automatic.
- Only three indicators: stress also depends on workload, diet, exercise, social support and more, none of which the system considers.
- One set of thresholds for everyone: normal HRV differs with age, fitness and personal baseline.
- Local accounts: the registration and login are handled locally in the prototype. They are fine for a demonstration but are not a replacement for a secure server-side system.
- Testing: the automated tests cover the Python version of the fuzzy and recommendation logic. A wider usability study with real users was limited by time.

The biggest strength of the system is transparency. Every result can be traced to membership values and rules, which many black-box detectors cannot offer. Separating the classifier from the recommendation engine also worked well. On the other hand, the quality of the output depends entirely on the quality of the rules and parameters, which is the usual trade-off for knowledge-based systems. In future we would like to connect the tool to a wearable for automatic HRV readings, personalise the membership functions to each user's own baseline, tune the parameters with real data and expert help, add more indicators, and run a proper usability study with students and employees.

11. Ethical and Social Considerations

Because the system deals with health and well-being, we thought about several ethical points.

- Not a medical tool: stressina.AI gives educational estimates only. It does not diagnose any condition and does not replace professional advice, and the interface carries a disclaimer saying so. Anyone who feels seriously distressed should contact a qualified professional or counsellor.
- Privacy: body and behaviour data are sensitive. The prototype asks for no personal identifiers beyond a local account, hashes passwords and provides privacy toggles. A

future version that connects to wearables or cloud services would need informed consent, secure storage and clear data policies.

- The companion: the Stressina mascot is meant to be supportive and calming. It should never be presented as a therapist, and it should point users to real help when they need it.
- Transparency and trust: by showing membership values and fired rules, the system lets users question a result instead of accepting it blindly.
- Over-reliance and anxiety: a number on a screen might worry some users or be trusted too much. The wording of results is gentle, and the recommendations are low-risk actions like breaks, breathing exercises and sleep habits.
- Fairness: thresholds that suit one group may not suit others, for example people of different ages or with conditions that affect HRV, so results should be seen as general guidance.
- Misuse: if an employer or institution used a tool like this, it should be voluntary and only for support, never for monitoring or judging individuals.
- Honest reporting: we have clearly said that the dataset is synthetic and that its labels come from the system's own pipeline, so that accuracy is not overstated.

On the positive side, a tool like this can help students and employees notice stress earlier and build healthier daily habits, and it can work alongside existing counselling and support services.

12. Conclusion

In this project we designed and built an explainable AI solution for everyday stress awareness. A Mamdani fuzzy inference system with 31 rules turns HRV, daily screen time and sleep duration into a Stress Index and a Low, Medium or High category, and a separate rule-based engine suggests a suitable coping action. An explanation layer shows exactly how each result was reached. The idea started as a Python and Streamlit prototype and ended as stressina.AI, a redesigned web application with a Liquid Glass interface, a relaxation and breathing area, a panda companion, personal history charts and privacy controls.

All the Stage 1 objectives were achieved, and the system was checked with automated tests and a repeatable synthetic dataset. Because it depends on simulated data and expert-designed parameters, it should be seen as an educational prototype and not a clinical tool. With real data, wearable support and expert review, the same approach could grow into a more reliable and

personalised well-being assistant. The project also taught us a lot about fuzzy systems, software design, testing and working as a team.

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Individual Contributions

The table below summarises the contribution of each group member to the project.

Member	Reg. No	Primary Area	Contributions
RMPD Jayatahissa	7080	AI Logic Design	Finalised the fuzzy membership curves and parameters; defined and documented all 31 Mamdani rules; implemented the fuzzy variables.
UHB Marasinghe	D/DBA/25.0015	Backend & Model	Built the inference engine (fuzzification, rule evaluation and defuzzification); integrated the recommendation engine; wrote the core pipeline code and application entry point.
JADD Jayawickrama	D/BIT/24/0020	UI & Requirements	Implemented the dashboard and visualisations; carried out manual scenario walkthroughs using the High, Medium and Low presets.
AA Madushavi	D/BIS/24/0041	Data & Systems	Generated and documented the synthetic dataset; wrote the automated pytest suite; compiled the references and the project documentation.

Table 6. Individual contributions of group members